

Case: Support Ticket Classification FineTuning Using Gen AI (OpenAI LLM)

Problem Statement:

Customer support teams often handle high volumes of incoming support tickets, requiring manual categorization and prioritization. This process can be time-consuming, prone to human error, and inconsistent, which impacts response times and customer satisfaction. Existing rule-based classification systems struggle to adapt to diverse ticket contents, making it difficult to accurately identify ticket topics, urgencies, or required expertise. As ticket volumes grow, manual intervention becomes unsustainable, leading to inefficiencies in handling customer inquiries and potential delays in resolution.

Goal:

The objective of this project is to demonstrate how Generative AI, specifically OpenAI's GPT models, can be applied to automatically classify support tickets. Automated classification helps streamline ticket routing, reduce manual workload, and improve response accuracy. This approach supports customer service teams by enabling faster, more consistent ticket categorization, ultimately enhancing customer satisfaction and operational efficiency.

Approach:

Generative AI models are used to analyze support ticket content, automatically assessing:

By leveraging AI, support tickets can be categorized based on unstructured content, allowing the system to "understand" ticket themes and apply consistent categorization criteria. This AI-driven approach reduces manual intervention and increases the scalability and accuracy of customer support workflows.

In this evaluation, the fine-tuned model's identifier is specified, and the same functions are employed to measure accuracy on both datasets.

Results and Analysis

The fine-tuned model demonstrated improved accuracy compared to the base model, indicating that fine-tuning effectively enhances the model's capability to classify support tickets accurately.

Conclusion

Fine-tuning the GPT-4o-mini-2024-07-18 model with task-specific data leads to notable performance improvements in support ticket classification. This process tailors the model to

better understand and respond to domain-specific inputs, thereby increasing its utility in specialized application

Conclusion:

The accuracy improved from the base model from 76% to 98% with 30000 tokens and classification labeled data and cost of training in 10 cents only!

Code snippets:

```
1 # Determine how the base model without any fine-tuning performs
2 model_id = 'gpt-4o-mini-2024-07-18'
3
4 training_responses = classify_tickets(
5     tickets=training_tickets,
6     model=model_id
7 )
8 accuracy = evaluate_accuracy(training_responses, training_labels)
9 print(f"Training Set Accuracy: {accuracy}%")
10
11 test_responses = classify_tickets(
12     tickets=test_tickets,
13     model=model_id
14 )
15
16 accuracy = evaluate_accuracy(test_responses, test_labels)
17 print(f"Test Set Accuracy: {accuracy}%")
```

Training Set Accuracy: 72.06%
Test Set Accuracy: 67.65%

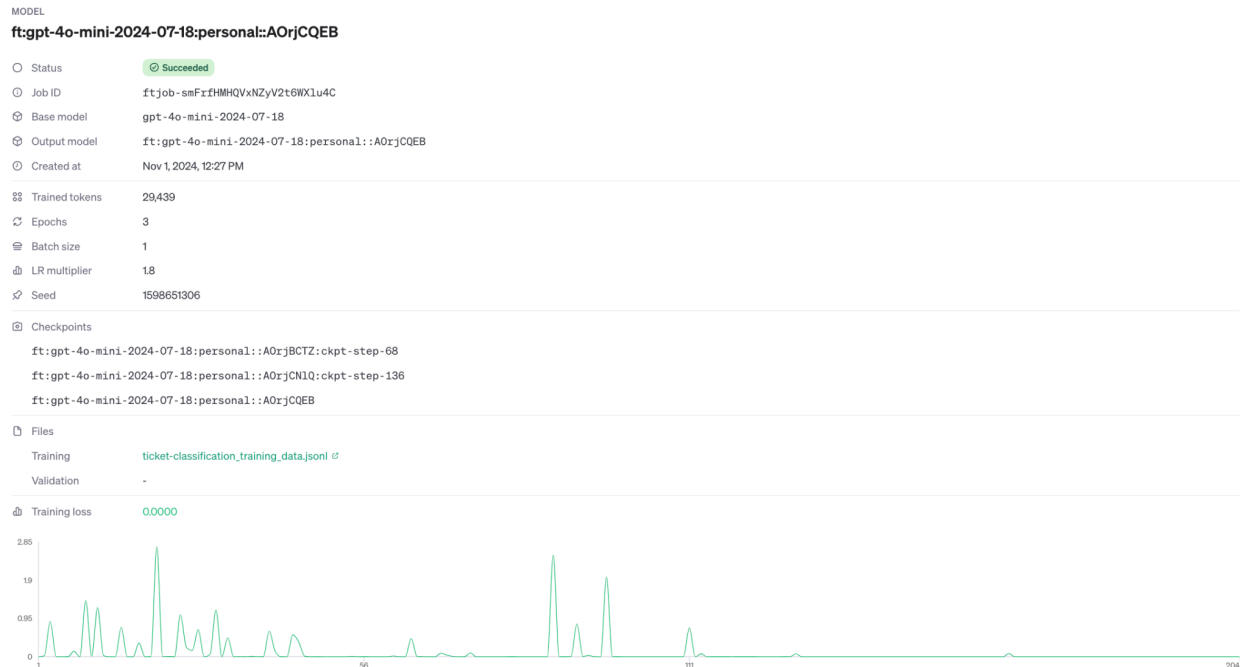
Base Model

```
1 # Determine how the base model performs with the increases rank, epochs, and learning
2 # https://platform.openai.com/finetune/ftjob-smFrHMHQVxNZyV2t6WXlu4C?filter=all
3 # REPLACE THIS WITH THE OUTPUT MODEL ID IN THE OPENAI FINE-TUNING DASHBOARD
4 model_id = 'ft:gpt-4o-mini-2024-07-18:personal::A0rjCQEBXYZ'
5
6 training_responses = classify_tickets(
7     tickets=training_tickets,
8     model=model_id
9 )
10 accuracy = evaluate_accuracy(training_responses, training_labels)
11 print(f"Training Set Accuracy: {accuracy}%")
12
13 test_responses = classify_tickets(
14     tickets=test_tickets,
15     model=model_id
16 )
17
18 accuracy = evaluate_accuracy(test_responses, test_labels)
19 print(f"Test Set Accuracy: {accuracy}%")
```

Training Set Accuracy: 98.53%
Test Set Accuracy: 97.06%

Fine Tuned Model

Model Metrics



About Author

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Senior Salesforce (SFDC), AI, and CRM Program Architect, highly skilled in integrating cutting-edge technologies like artificial intelligence and customer relationship management (CRM) platforms. With over **20 years of IT experience** (including 12 years in CRM/Salesforce and 5 years in AI), the author is a recognized leader in designing and delivering innovative AI and CRM solutions across industries. They have extensive expertise in **security, multi-org setups, data integration, design patterns, DevOps, and AI strategy**, and hold **20 Salesforce and 5 AI certifications**.

Feel free to reachout me for any AI/CRM free suggestions

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